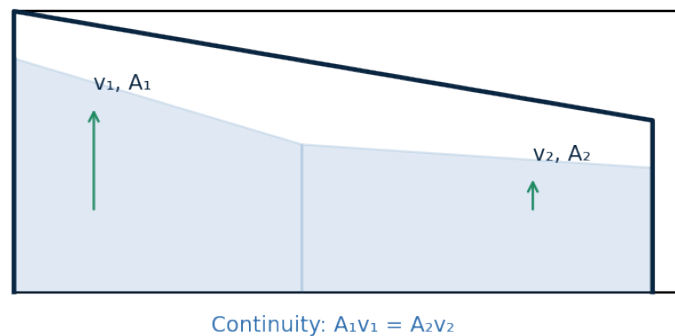


Properties of Solids and Fluids

JEE Physics — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- Stress = force per unit area; strain = fractional deformation. Young's modulus Y = stress/strain for longitudinal deformation (solids).
- Hooke's Law: stress is proportional to strain within the elastic limit.
- Pressure in a fluid at depth h : $P = P_0 + (\rho)gh$. Pascal's Law: pressure applied to an enclosed fluid is transmitted equally in all directions.
- Archimedes' Principle: the buoyant force on a submerged body equals the weight of the fluid displaced.
- Equation of continuity (incompressible, steady flow): $A_1 v_1 = A_2 v_2$ (mass conservation).
- Bernoulli's equation: $P + (1/2) \rho v^2 + \rho g h = \text{constant}$ along a streamline (energy conservation for ideal fluid flow).
- Surface tension is the force per unit length acting along the surface of a liquid; capillary rise $h = \frac{2 T \cos(\theta)}{\rho g r}$.
- Viscosity opposes relative motion between fluid layers; Stokes' law gives the viscous drag on a small sphere: $F = 6 \pi \eta r v$.



Fluid flow through a tube of varying cross-section, illustrating the equation of continuity.

2. Formula Sheet

Young's modulus	$Y = (F/A) / (\Delta L / L)$
Pressure at depth h	$P = P_0 + \rho g h$
Equation of continuity	$A_1 v_1 = A_2 v_2$
Bernoulli's equation	$P + (1/2) \rho v^2 + \rho g h = \text{const}$
Buoyant force (Archimedes)	$F_B = \rho(\text{fluid}) g V(\text{displaced})$
Capillary rise	$h = 2 T \cos(\theta) / (\rho g r)$
Stokes' law (viscous drag)	$F = 6 \pi \eta r v$
Terminal velocity of a sphere	$v_t = 2r^2(\rho - \sigma)g / (9 \eta)$

3. Practice Problems (22 questions)

1. A wire of length 2 m and cross-sectional area 1 mm^2 is stretched by 0.5 mm under a force of 100 N. Find its Young's modulus.
2. Find the pressure at a depth of 10 m below the surface of water ($\rho = 1000 \text{ kg/m}^3$, atmospheric pressure = $1.0 \times 10^5 \text{ Pa}$, $g = 10 \text{ m/s}^2$).
3. A block of wood of density 600 kg/m^3 floats on water (density 1000 kg/m^3). Find the fraction of its volume submerged.
4. Water flows through a pipe of cross-sectional area 4 cm^2 at 2 m/s. Find the speed of water in a narrower section of area 2 cm^2 .
5. State Bernoulli's principle and give one real-life application (e.g., aircraft wing lift).
6. A capillary tube of radius 0.5 mm is dipped in water ($T = 0.07 \text{ N/m}$, density 1000 kg/m^3 , $\theta = 0$). Find the height to which water rises ($g = 10 \text{ m/s}^2$).
7. A small steel ball of radius 1 mm falls through glycerine and attains a terminal velocity. State the three forces acting on the ball at terminal velocity.
8. Find the buoyant force on a fully submerged object of volume 0.002 m^3 in water (density 1000 kg/m^3 , $g = 10 \text{ m/s}^2$).
9. A hydraulic lift has a small piston of area 10 cm^2 and a large piston of area 200 cm^2 . Find the force needed on the small piston to lift a load of 2000 N on the large piston.
10. Explain why a ship made of iron floats on water while a solid iron ball sinks.
11. A metal sphere of density 8000 kg/m^3 and volume 0.001 m^3 is fully submerged in water. Find the apparent weight of the sphere in water ($g = 10 \text{ m/s}^2$).
12. State the two conditions required for Bernoulli's equation to be applicable to a fluid flow.
13. A wire is stretched by a force such that its length increases by 0.1%. If Young's modulus is $2 \times 10^{11} \text{ Pa}$, find the stress in the wire.

14. Find the excess pressure inside a soap bubble of radius 1 cm and surface tension 0.03 N/m (note: soap bubble has two surfaces).
15. A liquid rises to a height of 4 cm in a capillary tube of radius r . Find the height to which it would rise in a capillary tube of radius $r/2$ (same liquid).
16. State Pascal's law and give one everyday application.
17. Define the coefficient of viscosity and state its SI unit.
18. A piece of ice floats in a glass of water with a small pebble embedded in it. When the ice melts, does the water level rise, fall, or stay the same? Explain.
19. A liquid of density 1200 kg/m^3 flows through a horizontal pipe. At a point where the pipe narrows, the speed increases from 1 m/s to 4 m/s. Find the drop in pressure between the two points.
20. State Hooke's law and the meaning of the elastic limit.
21. A wooden cube of side 10 cm and density 800 kg/m^3 floats in water. Find the depth to which it sinks into the water.
22. Explain, using the equation of continuity, why water flows faster when a garden hose nozzle is narrowed.

Answer Key

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|---|---|
| 1. $Y = 4 \times 10^{11} \text{ Pa}$ | 2. $P = 2.0 \times 10^5 \text{ Pa}$ |
| 3. Fraction submerged = 0.6 (60%) | 4. $v_2 = 4 \text{ m/s}$ |
| 5. Conceptual - faster flow over the wing's curved top lowers pressure, generating lift | 6. $h = 2.86 \text{ cm}$ (approx) |
| 7. Conceptual - weight (down), buoyant force (up), viscous drag (up, opposing motion) | 8. $F_B = 20 \text{ N}$ |
| 9. $F = 100 \text{ N}$ | 10. Conceptual - average density of the ship's hollow hull is less than water; solid ball's density exceeds water's |
| 11. Apparent weight = 70 N | 12. Conceptual - flow must be steady, non-viscous, incompressible, and irrotational |
| 13. Stress = $2 \times 10^8 \text{ Pa}$ | 14. Excess pressure = 24 Pa |
| 15. $h' = 8 \text{ cm}$ (inversely proportional to radius) | 16. Conceptual - pressure applied to enclosed fluid transmits equally in all directions; e.g., hydraulic brakes |
| 17. Conceptual - η , SI unit Pa s (or N s/m^2) | 18. Conceptual - water level stays the same (floating ice displaces its own melted-water volume) |
| 19. Pressure drop = 9000 Pa | 20. Conceptual - stress proportional to strain up to the elastic limit; beyond it, permanent deformation occurs |
| 21. Depth submerged = 8 cm | 22. Conceptual - same flow rate ($A_1v_1 = A_2v_2$) through smaller area requires higher speed |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.